THE LINEAR MAZE

I. "CHOICE-POINT EXPECTANCY," "CORRECTNESS," AND THE GOAL-GRADIENT

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The question of identifying the factors affecting the serial order of difficulty of the different blinds in the maze has long been an intriguing problem. Apparently one of the first of such factors to be considered, historically, and one of the chief ones to be considered in the present report is that of the nearness of a given blind in serial order to the goal-box. Spence (2) has recently presented an excellent review of the literature for this particular problem in his study of the order of eliminating blinds in maze learning by the rat. It would be superfluous repetition to cite this literature here.

It is obvious that the maze pattern used in studying the serial order of blind elimination should control such differentially operating factors as directional trends in the maze, asymmetrically placed food position, extra-maze cues, heterogeneous sequences of turns and choices, and probably many other factors, which would tend to introduce differential errors independently of the experimental factor of mere proximity of the reward to individual blinds.

In order to construct a maze which would control these factors Warden (4) designed what he has called a symmetrical linear maze. This maze, made up of standard Warner-Warden units is intended to present a simple succession of identical elements, except for the fact of alternating right and left turns.

The present author in a preliminary report (1) presented evidence that Warden's maze does not present identical elements. Spence (2) has also criticized Warden's maze as having an asymmetrical factor.

The purpose of this experiment is to indicate some of the factors determining the order of error frequency in the *linear* maze.

APPARATUS

The maze used in this experiment is a modified form of the Warden linear maze. The modified T-maze form illustrated in

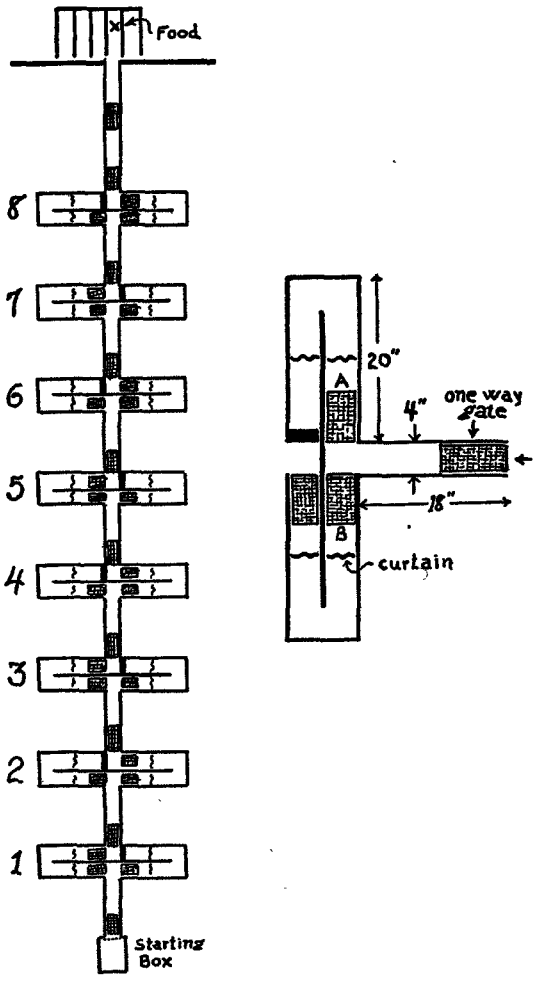


FIG. 1. MAZE PATTERN FOR GROUP I AND DETAIL OF A MAZE UNIT

figure 1 was used since each deviation of a true-path or blind from the central direction of the maze contains a compensating return. One of the units is shown in detail in the figure.

The rat enters the unit over a one-way gate made of hardware cloth. This gate is hinged at the bottom and rests normally at an angle of 45°. It is held in position by a rubber band. As the rat steps on to the gate it lowers flat to the floor. When the rat steps off the gate resumes its former position.

In both alleys at the choice-point are wire gates, *A* and *B* (in fig. 1), fastened flat to the floor. These gates were not used in this experiment but were in each unit for reasons that do not concern this problem.

The wriggly lines across the alleys are black curtains to prevent visual exploration. The right or left side of each unit could be made correct or incorrect by raising or lowering hardware cloth gates at the exit of a unit. In figure 1 the gate on the left at the exit of the unit detail is down, thus making this path correct, while the gate at the right is up, blocking the path and making that side of the unit a blind. The dimensions are as given in the figure. The maze was constructed of unpainted redwood.

Eight units constituted the maze. Food was placed 51 inches from the exit of the last unit.

It will be noted that this maze pattern attempts to overcome two asymmetrical features of the Warden maze. First, the alleys of each unit contain compensating returns to the general direction of the maze; second, the food position is centrally located with reference to the center of the maze. In the Y form (Warden) the food is at one side of the maze due to the Y-angle formed by the choice alleys and the stems of each unit. Units are numbered consecutively from starting box to food.

Lighting was effected by six frosted ceiling bowls, symmetrically placed on both sides of the maze, three to a side.

The animal living cages were directly behind the food box and about 5 feet from it. The walls of the room were uniformly white. White shades were drawn over the windows.

ANIMALS

All the animals used in this experiment were albino, female rats from the colony maintained by the California Psychological Laboratory. At the beginning of the experiment all animals were

about three months old. Due to the long learning period of 225 days required in the experiment, age might be thought to have some influence upon the results. Tryon (3) found no correlation between the ages of 141 rats and maze ability when the age range of the rats was from 100 days to 339 days. From these results it is highly probable that age is not an effective variable in this experiment.

PRELIMINARY TRAINING

All animals were given preliminary training in a straight alley 10 feet long. Curtains and gates were introduced gradually. The rats were given two runs per day for twelve days on this maze-familiarizing set-up.

EXPERIMENTAL PROCEDURE

There were two groups of animals, Group I, 14 animals, Group II, 13 animals. Both groups ran an alternation sequence of true-path choices. Each group ran a mirrored pattern of each other, Group I, LRLRLRLR, Group II, RLRLRLRL sequences of true-path turns.

The order in which the groups ran the maze was alternated from day to day so that if Group I ran first one day, Group II ran first the next day, thus giving each group an equal opportunity to run a "smelled-up" maze. Both groups were given one trial per day. The reward was an unlimited portion of Steenbock mash which the rats were allowed to eat for twelve minutes at the termination of each trial. Fresh lettuce was given to the animals once or twice a week in the living cages. All runs were made in the evening.

RESULTS

The error scores used in the data of this report are based upon all entrances into a blind beyond the first curtain adjacent to the choice-point. Retracing within a unit is not included, thus, the maximum error per unit per rat per trial is one error. Retracing from one unit to another is, of course, impossible due to the one-way gates between units.

In all the graphs and tables, the first day's run is not included.

The first twenty trials. The learning curve of each group for the first twenty trials is indicated in figure 2. The ordinate represents the average number of errors per day per rat per blind by two-trial periods. It will be noted that no learning has taken place and that both groups are comparable.

During this period, twenty days, it was found that the rats consistently made errors on one side of the maze, Group I to the right, Group II to the left. It will be noted in the maze diagram of figure 1 which shows the set-up for Group I that the last unit's

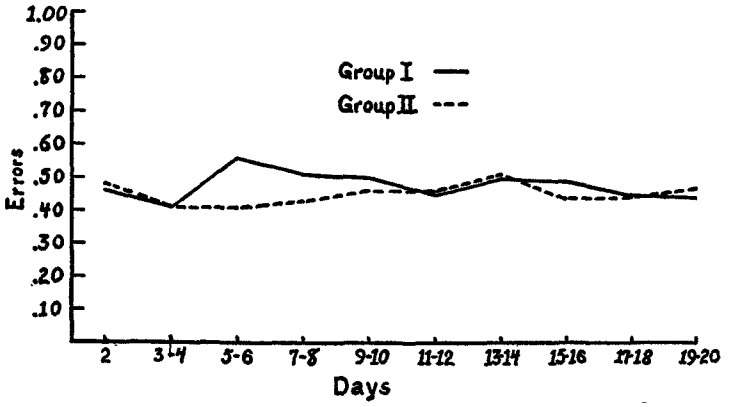


FIG. 2. AVERAGE ERROR CURVES FOR GROUP I AND GROUP II FOR THE FIRST TWENTY TRIALS

The ordinate represents errors per day per rat per blind per trial, the abscissa, trials by two-day periods.

(unit 8) correct choice is to the right. All the *blinds* on that side are odd numbered. Blinds on the opposite side of the maze are even numbered. Group II has the same situation, except that the last unit's correct choice is to the left. The *blinds* on that side are, of course, also odd numbered.

If we now plot the curves of average errors per rat per day per blind for the odd and even blinds we have the curves represented in figure 3. Since both groups behave identically, or nearly so, over this period and are running comparable patterns they have been combined in the curves.

The odd numbered blinds (pointing in the direction of the last

correct choice) show a high frequency of errors; the even numbered blinds (pointing away from the direction of the last correct choice) a low frequency.

It seems evident that we are here dealing with something analogous to a so-called position habit, and that it may possibly be caused by the last correct choice. To check this, on day 20, unit 8, the last in the maze, was removed. No other change was made in the maze. It is apparent that the removal of unit 8 places the correct final unit's choice, now in unit 7, on the opposite side of

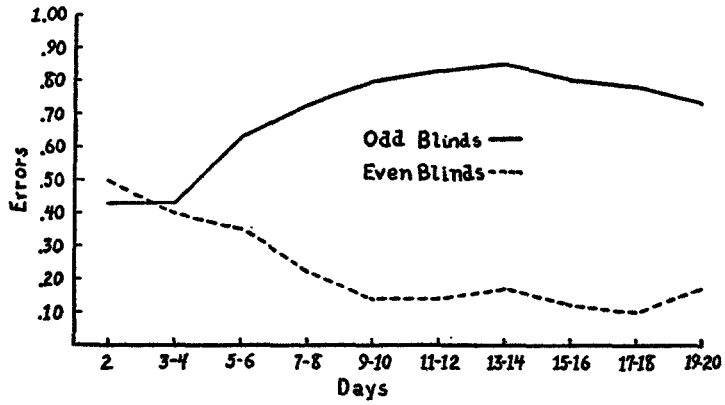


FIG. 3. AVERAGE ERROR CURVE OF ODD BLINDS (BLINDS IN THE SAME DIRECTION AS THE LAST CORRECT CHOICE) AND EVEN BLINDS (BLINDS IN THE OPPOSITE DIRECTION TO THE LAST CORRECT CHOICE) FOR BOTH GROUPS

The ordinate represents the average error per day per rat per blind; the abscissa the first twenty trials by two-day periods.

the maze to that of the 8 unit maze. In other words, the removal of unit 8, while shortening the maze by one unit, reverses for both groups the *side* into which a correct choice in the *last* unit is made.

Figure 4 gives the curves of the odd and even blinds for both groups for the full 225 days. Each point on the Y axis represents the average error per rat per blind per day for each 10 days on the X axis. The removal of unit 8 leaves 3 even numbered blinds and 4 odd numbered. All curves of odd and even blinds are, therefore, based upon data divided by the number of blinds involved.

The results of the removal of unit 8 on day 20 are indicated by the curves of figure 4. There is a reversal of errors from one side of the maze to the other though the separation of the odd blinds (now pointing away from the direction of the last correct choice) and even blinds (now pointing in the direction of the last correct choice) is not as marked as in the first 20 trials.

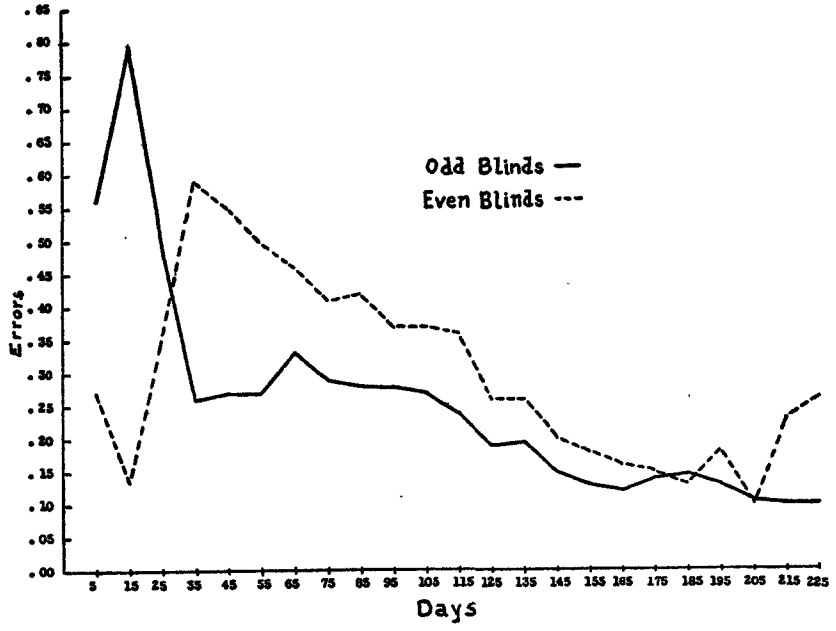


FIG. 4. AVERAGE ERROR CURVES OF ODD AND EVEN BLINDS FOR BOTH GROUPS FOR THE FULL EXPERIMENTAL PERIOD

The ordinate represents the average number of errors per day per rat per unit, the abscissa the trials for 225 days by ten-day periods. The days on the abscissa are the mid-points of each ten-day period.

It is now apparent that the last correct choice of the maze has operated to build up errors on *its* side of the maze.

In this maze the effect cannot be due to the position of the reward at one side or the other of the maze since its position has remained constant and centrally located with reference to the center of the maze direction.

Warden and Cummings (5), investigating primacy and recency factors in the linear maze say,

When we turn to the longer patterns we find no consistent tendency to order of elimination correlating with serial position, but, as in the Carr maze and Warden maze, a hit or miss order. This would lead us to conclude that neither primacy and recency nor any other principle operates in the maze type of serial motor learning, unless it can be shown that some factor entered into our results to disturb the normal order of fixation. For, we cannot criticize the apparatus: it presents a simple succession of identical elements, except for the fact of alternating right and left turns, and the data offer no evidence for supposing that a right turn is easier or harder than a left turn.

TABLE 1

Mean frequencies of entrances into culs-de-sac in the Warden linear maze. (From Warden and Cummings)

Culs-de-sac in order from entrance to goal

NUMBER OF CULS	PATTERN										TOTAL ODD CULS	TOTAL EVEN CULS
	1	2	3	4	5	6	7	8	9	10		
2	8.80	6.00									8.80	6.00
4	10.44	5.66	4.44	1.66							14.80	7.32
6	8.44	5.33	2.33	2.11	3.33	1.55					14.10	8.99
8	14.90	6.30	3.10	2.40	1.90	2.30	3.10	2.00			23.00	13.00
10	14.25	7.00	4.63	1.25	2.38	1.88	2.65	1.00	3.88	1.00	27.79	12.13
Totals											88.49	47.44

Ratio in favor of odd culs, 1.87.

Using Warden and Cummings' data a table has been compiled of mean error frequencies per rat for odd blinds and for even blinds. Since Warden and Cummings used an even number of units for all their mazes, the odd blinds always lead to the last correct choice side of the maze, while the even numbered blinds always branch off to the opposite side. We would expect a high frequency of errors on the odd numbered blinds and a relatively low frequency on the even numbered blinds. From table 1 it is evident that every length of maze used by Warden and Cummings was influenced by the last correct choice. When all units are combined, the odd score is 88.49, the even, 47.44 with a resulting

difference of 41.13. This gives a ratio in favor of the odd blinds of 1.87. Spence (2) has also pointed out the above factor, and has further shown that even though this factor is present, a general tendency for the latter part of the maze to be learned more readily than the first part can be shown in Warden and Cummings' results.

Since the maze used in the present experiment has equated the food position with respect to every part of the maze and since our results agree with Warden and Cummings it is highly probable that the last choice point is an intrinsic factor in linear mazes and cannot be experimentally controlled with the present forms of linear maze patterns.

TABLE 2

Average error per rat for each blind from day 31 to 225 inclusive, based on the twenty animals that ran the full experimental period

BLINDS IN ORDER FROM STARTING BOX TO MAZE EXIT						
1	2	3	4	5	6	7
57.55	51.20	50.80	60.30	30.35	72.75	25.80

The error curves of odd and even blinds, figure 4, indicate that the maximum establishment of the effect of the last correct choice, after the removal of unit 8 of the maze, is during the period from day 31 to day 40, i.e., about 15 days after the experimental change of the last correct choice. Therefore, the statistics to follow are concerned with data obtained from day 31 to day 225 inclusive.

The reason for considering this part of the learning is that we wish to indicate the serial order of error frequencies per blind in relation to the last correct choice of the maze.

Table 2 indicates the mean frequency per day per rat for each blind from day 31 to day 225.[1]

[1] These averages are based upon 20 rats. During the experiment 7 animals died at varying intervals (see table 3). These deaths were probably not entirely due to the rigors of the long experimental period (since the animals did not lose weight) but rather to the normal incidence of disease. All the curves and tables are based upon the full number of animals unless specific mention of the number is made. These data, table 2, were thought to be more significant if only those animals that lived until the end of the experiment were included.

It will be seen that, considering the full serial order, there is no consistent serial order of error frequency. Thus regarding all blinds the results agree with those of Warden and Cummings. But, if the odd blinds and the even blinds are considered independently of one another we are confronted with *two* consistent orders. Those blinds (even) which are in the direction of the last correct choice tend to have higher and higher error frequencies as they approach the last correct choice position of the maze. Those blinds (odd) that point away from the direction of the last

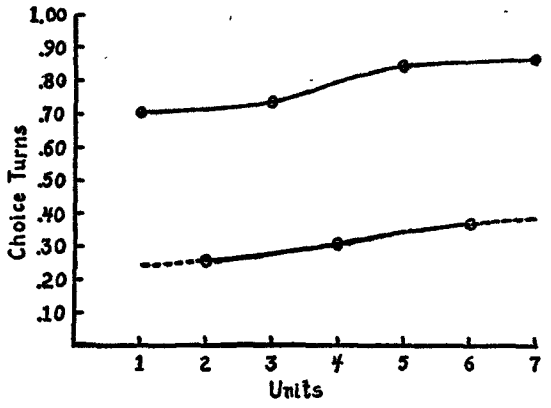


FIG. 5. AVERAGE NUMBER OF TURNS IN THE SAME DIRECTION AS THE LAST CORRECT CHOICE FOR EACH MAZE UNIT FROM DAY 31 TO 225 INCLUSIVE

The ordinate represents the average number of turns per day per rat for each unit, the abscissa the maze units from starting box to the maze exit. The dotted lines represent extrapolated portions of the lower curve.

correct choice tend to have fewer errors as they approach the last correct choice.

The explanation for the orders may be found the concept I originally termed "choice-point expectancy," and which has been elsewhere described as an "anticipatory" error (2). The "choice-point expectancy" hypothesis is based upon the similarity of all units with the last one of the maze. The correct choice in the last unit plus the expected reward is carried as a behavior unit to every other unit in the maze. As the linear distance from the last correct choice is increased the rat's "expectation" of a particular

choice being the last one in the maze decreases. In other words, all units are confused to some degree, depending upon the linear distance from the last choice, with the last unit of the maze, and the units by their similar structure support such behavior as is correct in the last unit. Thus, since all *turns* in the same direction as the last correct choice are made more frequently as the end of the maze is approached, those units having blinds in that direction increase in error frequency while those units which do not have *blinds* in that direction decrease in error frequency.

For a clearer exposition of this factor the analysis employed in the curves of figure 5 is used. The ordinate represents the average number of *turns* per unit per day per rat for days 31 to 225 inclusive.[2] The abscissa represents the maze units, unit 1 being the first, unit 7 being the last one in the maze. The upper curve is the curve for units 1, 3, 5, and 7. It will be remembered that for these units a turn in the same direction as the last correct choice would lead the animal into the *true* path. This upper curve, then, shows the average number of *correct choices* per unit per rat per day for these units. The lower curve represents the data for units 2, 4 and 6. Conversely, for these units a turn in the same direction as the last correct choice would lead the animal into a blind. This lower curve, then, shows the average number of *errors* per day per unit per rat.

Considering first the lower curve we see, as was shown above, that as the end of the maze is approached more and more turns are made in the same direction as the last correct choice. In other words, the *error* score actually becomes greater as the end of the maze is approached. This increase can only be explained on the assumption that the choice-point expectancy force increases at a definite rate as the end of the maze is approached. From a 27 per cent error score in blind 2 we get an increase to a 37 per cent error score in blind 6.[3]

Considering the upper curve we find that the rate of increase (i.e., slope of the upper curve) is almost identical with the lower

[2] Only the 20 rats that ran the full 225 days are included in data represented by these curves.

[3] The dotted lines of the lower curve, figure 5, are extrapolations.

curve. If we assume that the choice-point expectancy force was working for these units in the very same way and with the very same force as for units, 2, 4, and 6 (lower curve) we can see why these two curves should have the same slope, the changes in both curves being due to the same factor. However, if this factor were the only one acting upon both curves we should expect the two curves to be superimposed. Actually, as is apparent from the graph, these two curves while having the same slope run at different levels. This phenomenon of parallel curves could be easily explained if we would postulate a given *constant* force working throughout units 1, 3, 5, and 7, and not for 2, 4, and 6. With this set-up, i.e., one variable force affecting both curves in the same way and one constant force affecting only the upper curve the graph becomes intelligible. We have already pointed out what the variable force might be, i.e., the "choice-point expectancy force." We now suggest that the constant force working on units 1, 3, 5, and 7 is due to the fact that a turn in accordance with the "choice-point expectancy force" is also "correct." This would mean that the factor of correctness reënforces a given response (which response might have been first induced by the choice point expectancy force) *equally throughout the maze.* Correctness then does not at all obey the goal gradient hypothesis, but is a constant.

It might be objected that instead of starting our analysis with the lower curve had we started with the upper curve we would have come to the conclusion that correctness is the variable which obeys the goal gradient hypothesis and choice point expectancy is the constant factor. But this can easily be shown not to be the case. For if we assume that the positive slope of the upper curve is due to the fact that the potency of correctness is increased as the end of the maze is approached we would expect the lower curve to show a *negative slope* (i.e., a decrease in errors) as the end of the maze is approached. However, just the reverse is the case, both curves showing a positive slope.

These two hypotheses, first, that choice-point expectancy obeys the goal gradient principle and, second, that correctness does not obey the goal gradient principle but is constant throughout the

maze adequately explains the two consistent orders of difficulty for the odd and even blinds.

Thus, proximity in time and linear distance of a series of maze units to the final goal does not seem to differentially affect the value of "rightness" or "wrongness" of any choice behavior in a particular unit.

TABLE 3

Average number of errors per rat per day per blind for the periods and numbers of animals indicated

PERIOD	N	ODD BLINDS IN ORDER				EVEN BLINDS IN ORDER			
		1	3	5	7	2	4	6	8
2–10	27	0.659	0.467	0.585	0.585	0.207	0.296	0.319	0.267
11–20	27	0.729	0.796	0.852	0.874	0.130	0.159	0.159	0.020
21–30	26	0.388	0.473	0.538	0.604	0.369	0.392	0.288	
31–40	26	0.358	0.204	0.200	0.269	0.515	0.596	0.658	
41–50	26	0.458	0.200	0.189	0.246	0.462	0.658	0.577	
51–60	26	0.446	0.212	0.200	0.219	0.373	0.531	0.592	
61–70	26	0.430	0.277	0.285	0.231	0.339	0.542	0.515	
71–80	25	0.336	0.312	0.228	0.224	0.320	0.440	0.408	
81–90	24	0.392	0.304	0.246	0.163	0.308	0.463	0.367	
91–100	24	0.396	0.242	0.221	0.292	0.254	0.446	0.383	
101–110	24	0.413	0.271	0.175	0.196	0.279	0.413	0.404	
111–120	24	0.346	0.325	0.146	0.117	0.263	0.429	0.433	
121–130	24	0.321	0.246	0.125	0.088	0.221	0.217	0.329	
131–140	24	0.308	0.300	0.117	0.063	0.233	0.213	0.292	
141–150	24	0.208	0.204	0.113	0.046	0.183	0.146	0.233	
151–160	24	0.146	0.225	0.100	0.033	0.125	0.146	0.208	
161–170	24	0.183	0.204	0.071	0.046	0.129	0.117	0.208	
171–180	24	0.204	0.158	0.092	0.079	0.142	0.083	0.213	
31–80	26	0.406	0.241	0.220	0.238	0.402	0.553	0.550	
81–130	24	0.374	0.276	0.183	0.171	0.265	0.394	0.383	
131–180	24	0.210	0.218	0.099	0.053	0.162	0.141	0.231	
181–225	20	0.139	0.192	0.090	0.056	0.163	0.106	0.266	
31–225		0.281	0.232	0.148	0.129	0.273	0.300	0.356	

An examination of the mean frequencies for different stages of learning, table 3, indicates that, in general, the two orders of error frequency are consistent throughout the learning period. It will be noticed that the even blinds, those that point in the direction of the last correct choice, establish their order of error frequency

in the period from day 31 to 80, while those blinds which point away from the direction of the last correct choice establish their order in the period from day 81 to 130. During the first twenty days, these latter blinds were in the direction of the last correct choice, but (due to the removal of unit 8) are now in the opposite direction. The even blinds, or those that during the first twenty days pointed away from the direction of the last correct choice, now point in the direction of the last correct choice.

The difference in trials necessary to establish the new orders can be explained by examining the results of Yoshioka's (6) experiments on direction as a factor in maze solution. This experimenter found that if a position habit is established in the goal direction it is more difficult to break up and to establish a new habit in opposition to the goal direction than when the original habit is away from the goal direction and the new habit is in the direction of the goal. In our experiment those blinds that originally pointed in the direction the last correct choice (odd) now point away from the direction of the last correct choice, and hence present a more difficult problem than those blinds (even) that originally pointed away from the direction of the last correct choice and now point toward the last correct choice. Thus, in terms of the difficulty of establishing a new habit in opposition to the direction of the last correct choice we may explain the earlier establishment of the final error frequency order of those blinds that point in the *new* direction of the last correct choice.

SUMMARY

1. In an eight-unit linear maze the last correct choice before the food compartment was found to determine the relative difference in error frequencies between those blinds pointing in the same direction and those blinds pointing in the opposite direction to the last correct choice, more errors being made into the former than the latter blinds.

2. Warden and Cummings' results of a hit or miss order of error frequency for serial blind order in the linear maze are found to have been influenced by the effect of the last correct choice.

3. Two serial orders of error frequency are found: (*a*) a de-

creasing frequency as the end of the maze is approached for those blinds pointing away from the direction of the last correct choice, and (*b*) an increasing error frequency as the end of the maze is approached for those blinds that point in the same direction as the last correct choice. This behavior is given a theoretical explanation in the concept of "choice-point expectancy."

4. The average difference of the two orders may be explained by two factors: (*a*) the "choice-point expectancy" variable acting with *increasing* force as the last maze unit position is approached, and (*b*) the correctness of the choice, acting as a *constant* to increase the frequency of *correct* turns in the same direction as the last correct choice.

5. The difference in the rate of establishing a new error frequency order when the direction of the last correct choice has been reversed can be explained by the relative difference in difficulty of establishing a new habit in opposition to a goal-associated direction when the former habit was in the direction of the last correct choice and the establishment of a new habit in opposition to the goal-associated direction when the former habit was in the direction of the last correct choice. The latter conditions effect a later final order of error frequency than the former conditions.

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